SET-K

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B.SC.[ECS] - II Sem III Practical Examination, October 2026

VSC1: Based on DSC1-3 and DSC1-4

Date & Time: _____ Max Marks: 30

Instructions:

I) Attempt any one question from each section.

II) Each question carries 10 marks.

III) The Journal and Viva-Voce carry 5 marks.

Section-I: Based on DSC1-3 (Data Structures using C++)

Q.1 Write a program to implement IRD with its basic operations.

Q.2 Write a program that converts infix expression of postfix expression using stack.

Section-II: Based on DSC1-4 (Database Management Systems)

Q.3 Create a table book by applying proper constraints, insert at least 5 records and solve the following queries:

Field Name	Data Type	Constraint
Book_ID	INT	Primary Key
Book_Name	VARCHAR(100)	NOT NULL
Author	VARCHAR(50)	-
Publisher	VARCHAR(50)	-
Category	VARCHAR(30)	-
Price	DECIMAL(8,2)	-
Quantity	INT	-

a. Display all books available in the library.

b. Display the names and prices of all books.

c. Display books belonging to the 'Computer Science' category.

d. Display books whose price is greater than 500.

e. Display books whose quantity is less than 5.

Q.4 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

A] Customer:

Field Name	Data Type	Constraint
Customer_ID	INT	Primary Key
Customer_Name	VARCHAR(50)	NOT NULL
City	VARCHAR(30)	-
Phone	VARCHAR(15)	-

B] Order:

Field Name	Data Type	Constraint
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key referencing Customer(Customer_ID)
Order_Date	DATE	-
Amount	DECIMAL(10,2)	-
Status	VARCHAR(20)	-

a. Display customer names with their order IDs and amounts using an INNER JOIN.

b. Display all customers, including customers who have not placed orders, using a LEFT JOIN.

c. Display customers who have placed orders worth more than ₹5,000.

d. Display customers whose total order amount exceeds ₹10,000.

e. Display orders whose amounts are greater than the average order amount.